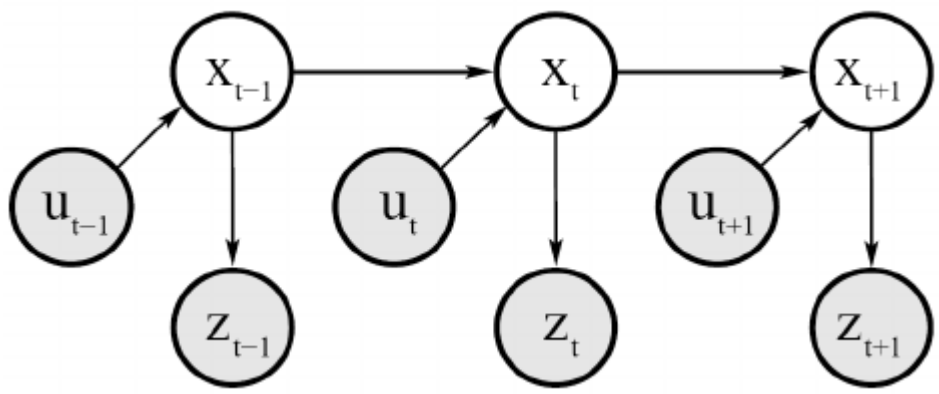
Ro5: Probabilistic Motion Models

- Modelling Motion Uncertainty

Remember: State Estimation of state x of a system given observation z and controls u

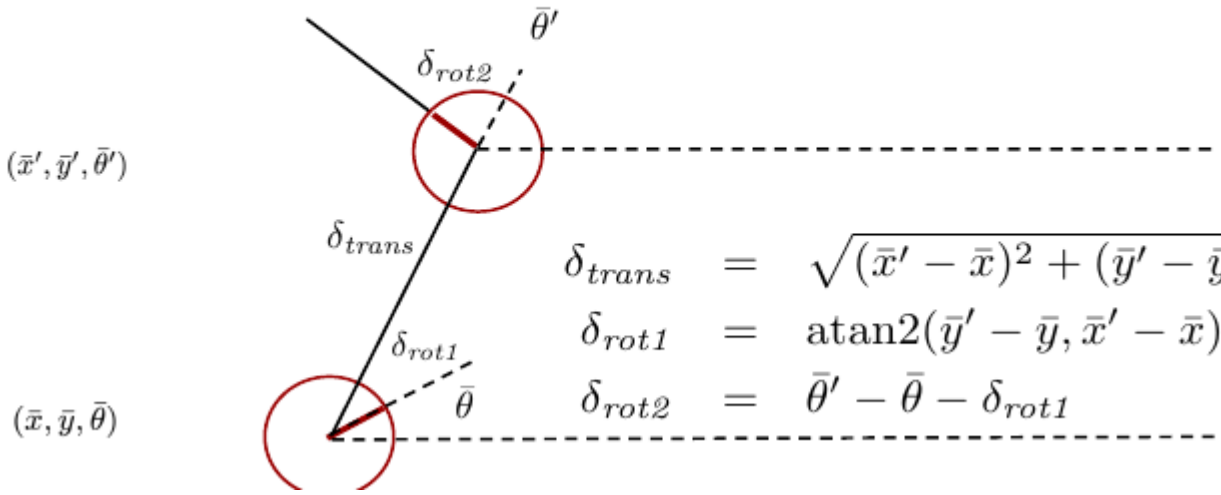
- $p(x_t)|u_t, x_{t-1})$
  - Probabilistic Motion Models
    - Specifies a posterior probability that action u carries the robot form x_t to $x_{t-1}
- Via Recursive Bayes Filter
  - **Prediction step**
  - $\overline{bel}(x_t) = \int \underbrace{p(x_t|x_{t-1}, u_t)}_{\text{motion model}} \cdot bel(x_{t-1}))dx_{t-1}$
  - **Correction step**
    - $bel(x_t) = \eta \cdot \underbrace{p(z_t|x_t)}_{\text{observation model}} \overline{bel}(x_t)$

Dynamic Bayesian Network for States, Controls, and Observations



Odometry Motion Model

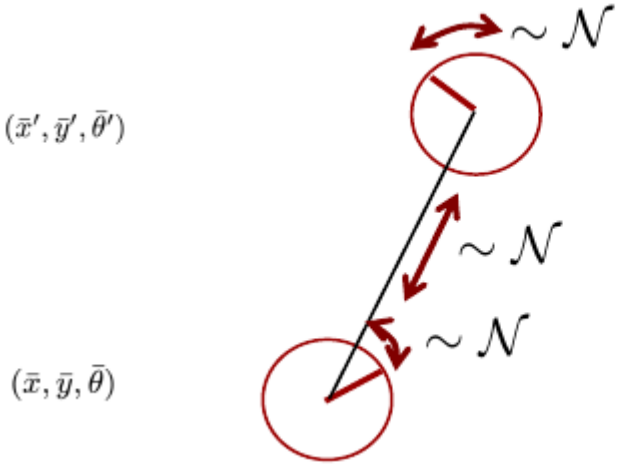
- Motion from $(\bar{x}, \bar{y}, \bar{\theta})$ to $(\bar{x}', \bar{y}', \bar{\theta}')$ with $u = (\delta_{rot1}, \delta_{trans}, \delta_{rot2})$



$$\begin{aligned} \delta_{trans} &= \sqrt{(\bar{x}' - \bar{x})^2 + (\bar{y}' - \bar{y})^2} \\ \delta_{rot1} &= \operatorname{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta} \\ \delta_{rot2} &= \bar{\theta}' - \bar{\theta} - \delta_{rot1} \end{aligned}$$

- Gaussian noise in the individual odometry parameters $u = (\delta_{rot1}, \delta_{trans}, \delta_{rot2})$

Noise Model



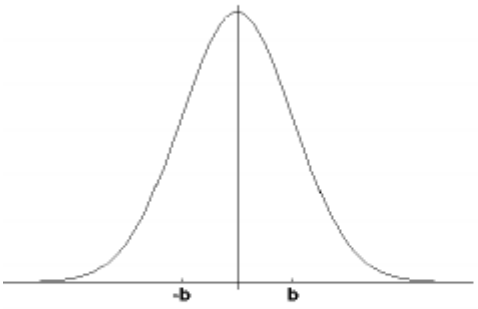
$$u \sim \mathcal{N}(0, \Sigma)$$

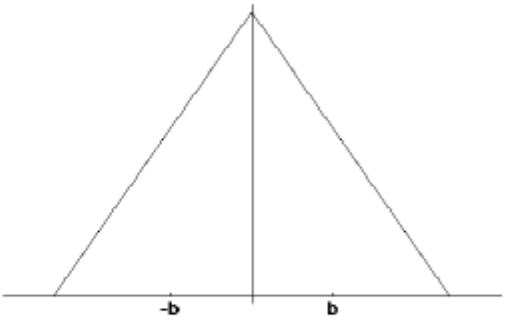
- Measured motion is true motion with noise

$$\begin{aligned} \hat{\delta}_{rot1} &= \delta_{rot1} + \epsilon_{\alpha_1|\delta_{rot1}|+\alpha_2|\delta_{trans}|} \\ \hat{\delta}_{trans} &= \delta_{trans} + \epsilon_{\alpha_3|\delta_{trans}|+\alpha_4(|\delta_{rot1}|+|\delta_{rot2}|)} \\ \hat{\delta}_{rot2} &= \delta_{rot2} + \epsilon_{\alpha_1|\delta_{rot2}|+\alpha_2|\delta_{trans}|} \end{aligned}$$

Typical Distributions

Normal distribution **Triangular distribution**





$$\varepsilon_{\sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{x^2}{\sigma^2}}$$

$$\varepsilon_{\sigma^2}(x) = \begin{cases} 0 \text{ if } |x| > \sqrt{6\sigma^2} \\ \dfrac{\sqrt{6\sigma^2} - |x|}{6\sigma^2} \end{cases}$$

Calculating the Posterior Given x, x', and Odometry

hypotheses odometry

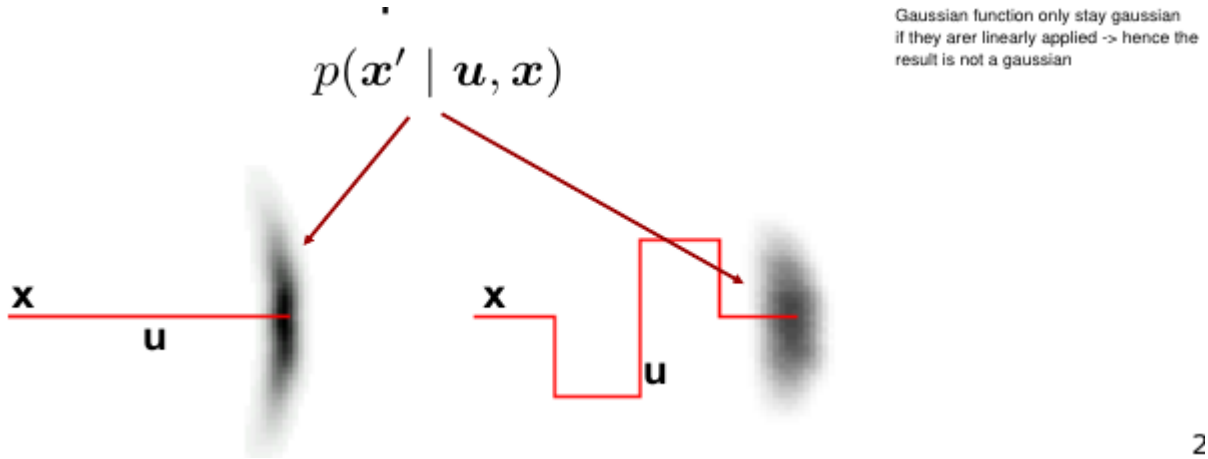
1. **Algorithm motion_model_odometry**$(\boxed{x, x'}, \boxed{\bar{x}, \bar{x}'})$
2. $\delta_{trans} = \sqrt{(\bar{x}'-\bar{x})^2 + (\bar{y}'-\bar{y})^2}$
3. $\delta_{rot1} = \text{atan2}(\bar{y}'-\bar{y}, \bar{x}'-\bar{x}) - \bar{\theta}$
4. $\delta_{rot2} = \bar{\theta}' - \bar{\theta} - \delta_{rot1}$
5. $\hat{\delta}_{trans} = \sqrt{(x'-x)^2 + (y'-y)^2}$
6. $\hat{\delta}_{rot1} = \text{atan2}(y'-y, x'-x) - \bar{\theta}$
7. $\hat{\delta}_{rot2} = \theta' - \theta - \hat{\delta}_{rot1}$
8. $p_1 = \text{prob}(\delta_{\text{rot1}} - \hat{\delta}_{\text{rot1}}, \alpha_1 |\delta_{\text{rot1}}| + \alpha_2 \delta_{\text{trans}})$
9. $p_2 = \text{prob}(\delta_{\text{trans}} - \hat{\delta}_{\text{trans}}, \alpha_3 \delta_{\text{trans}} + \alpha_4 (|\delta_{\text{rot1}}| + |\delta_{\text{rot2}}|))$
10. $p_3 = \text{prob}(\delta_{\text{rot2}} - \hat{\delta}_{\text{rot2}}, \alpha_1 |\delta_{\text{rot2}}| + \alpha_2 \delta_{\text{trans}})$
11. **return** $p_1 \cdot p_2 \cdot p_3$

odometry parameters (u)

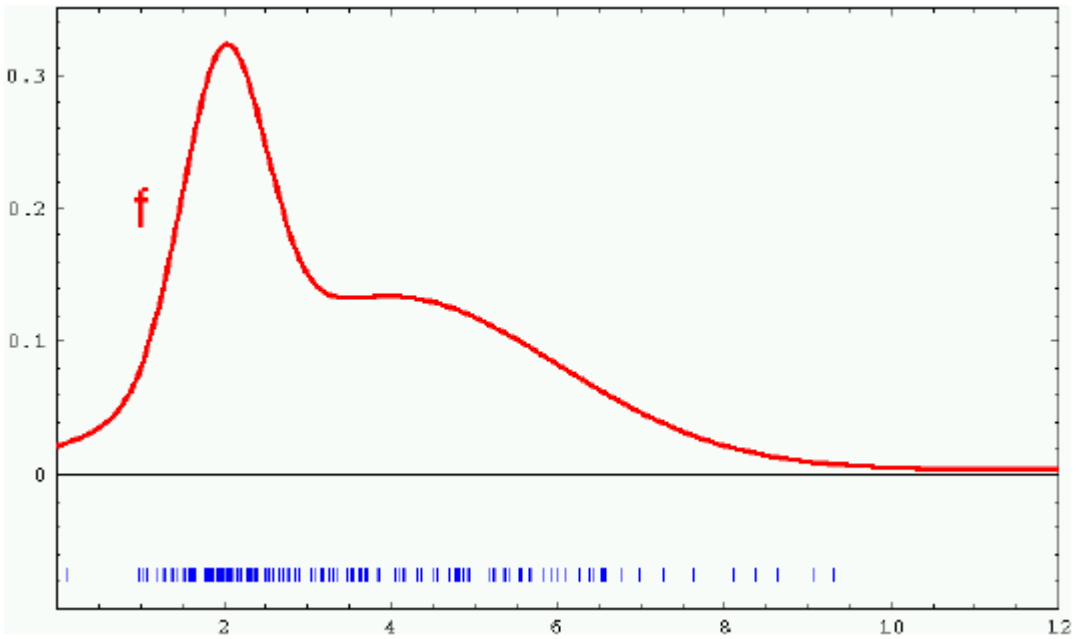
values of interest (x,x')

Application

- Repeated application of the motion model for short movements along the red trajectories
- Typical banana-shaped distributions obtained for the 2D-projection of the 3D posterios

- 

Sample-Based Density Representation

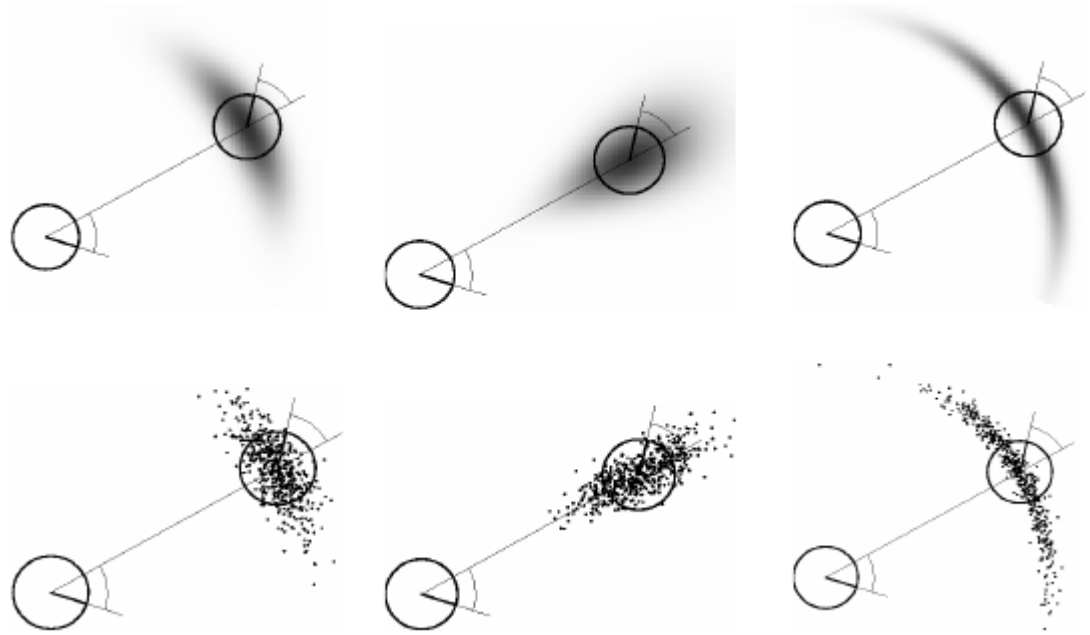


- Sampling from Normal or Triangular Distributions
  - $\frac{1}{2}\sum_{i=1}^{12} rand(-b, b)$
- Sample Odometry Motion Model
  - **Algorithm sample_motion_model(u, x):**

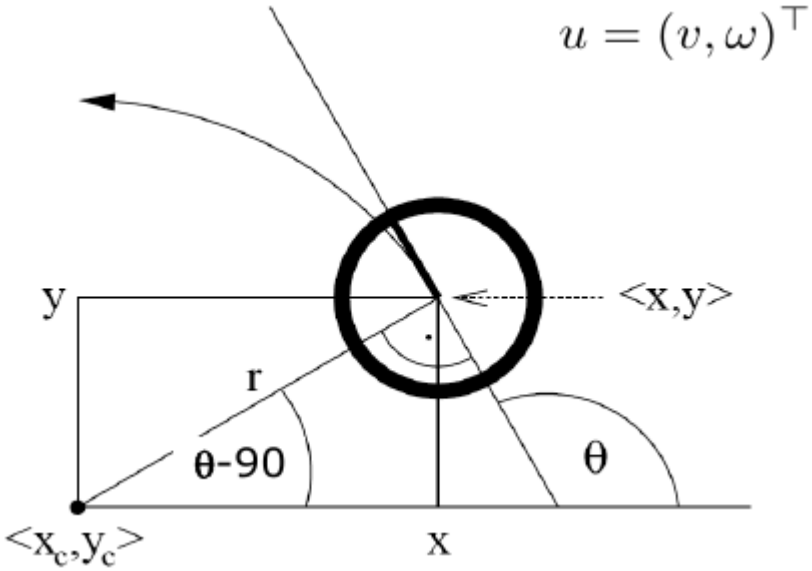
    $u = \langle \delta_{rot1}, \delta_{rot2}, \delta_{trans} \rangle, x = \langle x, y, \theta \rangle$

    1. $\hat{\delta}_{rot1} = \delta_{rot1} + \text{sample}(\alpha_1 \mid \delta_{rot1} \mid + \alpha_2 \ \delta_{trans})$
    2. $\hat{\delta}_{trans} = \delta_{trans} + \text{sample}(\alpha_3 \ \delta_{trans} + \alpha_4 \ (\mid \delta_{rot1} \mid + \mid \delta_{rot2} \mid))$
    3. $\hat{\delta}_{rot2} = \delta_{rot2} + \text{sample}(\alpha_1 \mid \delta_{rot2} \mid + \alpha_2 \ \delta_{trans})$
    4. $x' = x + \hat{\delta}_{trans} \cos(\theta + \hat{\delta}_{rot1})$
    5. $y' = y + \hat{\delta}_{trans} \sin(\theta + \hat{\delta}_{rot1})$
    6. $\theta' = \theta + \hat{\delta}_{rot1} + \hat{\delta}_{rot2}$
    7. **Return** $\langle x', y', \theta' \rangle$

    sample_normal_distribution

- 

Velocity-Based Model



- Noise Model: True motion corrupted with noise
  - $\hat{\nu} = \nu + \epsilon_{\alpha_1|\nu|+\alpha_2|\omega|}$
  - $\hat{\omega} = \omega + epsilon_{\alpha_3|\nu|+\alpha_4|\omega|}$
  - $\hat{\gamma} = \epsilon_{\alpha_5|\nu|+\alpha_6|\omega|}$
    - Term to account for final rotation
    - KaTeX parse error: Undefined control sequence: \nut at position 6: (\hat\nut,\hat\omega) circle constrains final orientation
      - 2D manifold in 3D space

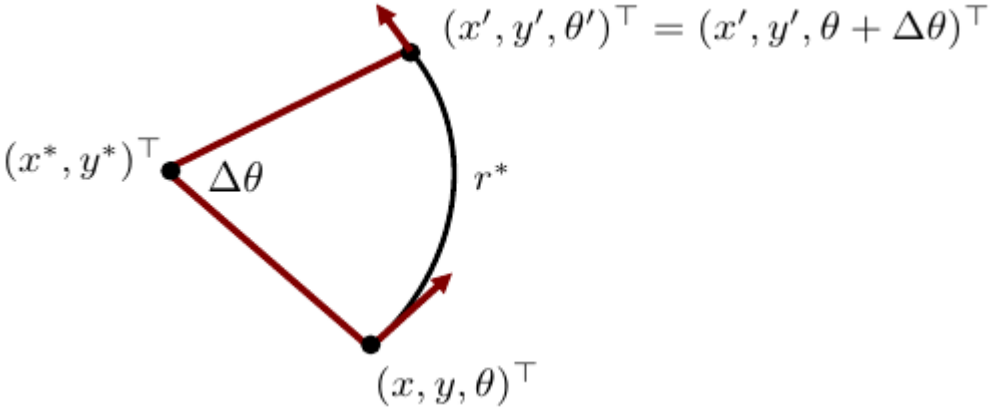
Equation

$x_{t-1} = (x, y, \theta)^\top$
$x_t = (x', y', \theta')^\top$
$u = (v, \omega)^\top = ?$

$$\begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} \frac{x+x'}{2} + \mu(y - y') \\ \frac{y+y'}{2} + \mu(x' - x) \end{pmatrix} \qquad \mu = \frac{1}{2} \frac{(x - x')\cos\theta + (y - y')\sin\theta}{(y - y')\cos\theta - (x - x')\sin\theta}$$

Thus, we can obtain the result

$$r^* = \sqrt{(x^* - x)^2 + (y^* - y)^2}$$
$$\Delta\theta = \operatorname{atan2}(y' - y^*, x' - x^*) - \operatorname{atan2}(y - y^*, x - x^*)$$



- The parameters of the circle:

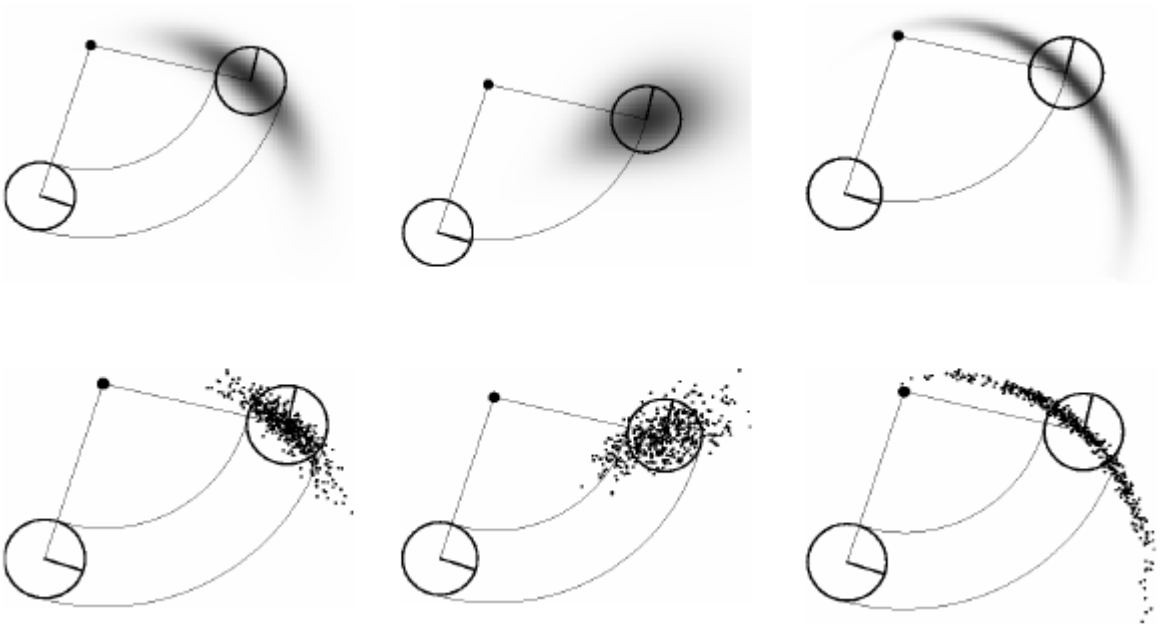
$$r^* = \sqrt{(x^* - x)^2 + (y^* - y)^2}$$
$$\Delta\theta = \operatorname{atan2}(y' - y^*, x' - x^*) - \operatorname{atan2}(y - y^*, x - x^*)$$

- allow for computing the velocities as

$$v = \frac{\Delta\theta}{\Delta t} r^*$$
$$\omega = \frac{\Delta\theta}{\Delta t}$$

Algorithm sample_motion_model_velocity(u_t, x_{t-1}):

$$\hat{v} = v + \textbf{sample}(\alpha_1 v^2 + \alpha_2 \omega^2)$$
$$\hat{\omega} = \omega + \textbf{sample}(\alpha_3 v^2 + \alpha_4 \omega^2)$$
$$\hat{\gamma} = \textbf{sample}(\alpha_5 v^2 + \alpha_6 \omega^2)$$
$$x' = x - \frac{\hat{v}}{\hat{\omega}} \sin\theta + \frac{\hat{v}}{\hat{\omega}}\ sin(\theta + \hat{\omega}\Delta t)$$
$$y' = y + \frac{\hat{v}}{\hat{\omega}} \cos\theta - \frac{\hat{v}}{\hat{\omega}}\ cos(\theta + \hat{\omega}\Delta t)$$
$$\theta' = \theta + \hat{\omega}\Delta t + \hat{\gamma}\Delta t$$
$$\textit{return}\ x_t = (x', y', \theta')^T$$



Summary

- odometry-base and velocity-based motion models
- Calculating posterior probability $p(x_t|u_t, x_{t-1})$
- Sampling from $p(x_t|u_t, x_{t-1})$
- Samplingn from $p(x_t|u_t, x_{t-1})$
- Typically in intervals Δt between t-1 and t
- Model parameters have to be learnt/estimated